MILITARY SPECIFICATION SHEET

PAINT, EPOXY-POLYAMIDE, ZINC PRIMER, FORMULA 159, TYPE II

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the paint described herein shall consist of this specification and the latest issue of MIL-P-24441(SH).

FORMULA. This formula covers zinc epoxy-polyamide paint designated by Navy formula 159, type II for interior or exterior use. The paint shall consist of the ingredients specified in the quantities specified.

Component A

	Pounds
Polyamide 2/	8
Polyamide adduct 3/	106
2-ethoxy ethanol 4/	24
2-butoxy ethanol 5/	7.6
Paint thinner, TT-T-291, type II, grade A	13.2

Component B

	Pounds
Zinc pigment 6/	1900
Epoxy resin 7/	190
Thixotrope 1/	15
2-ethoxy ethanol 4/	100.6
2-butoxy ethanol 5/	30.2
Paint thinner, TT-T-291, type II, grade A	58.7

See footnotes on next page.

AMSC N/A
FSC 8010
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

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Formula 159 thinner

	Pounds
2-ethoxy ethanol[4/]	387
2-butoxy ethanol[5/]	151
Paint thinner, TT-T-291, type II, grade A	198

QUANTITATIVE REQUIREMENTS. The paint shall meet the following quantitative requirements and the qualitative requirements of section 3 of the general specification. Components A and B shall be mixed 1:4 by volume for mixed components tests. Tests shall be performed in accordance with the general specification.

Requirements[8/]	Component A		Component B		Mixed component	
	Min	Max	Min	Max	Min	Max
Pigment content, percent	—	—	81.5	85.5	—	—
Volatiles, percent	42.8	46.8	8.1	8.5	—	—
Nonvolatile vehicle, percent	53.1	57.1	8.0	8.6	—	—
Weight per gallon, pounds	7.8	8.2	28.5	28.9	—	—
Water, percent	—	0.5	—	0.2	—	—
Flash point, °F	99	—	110	—	99	—
Sag resistance, mils	—	—	—	—	8	—
Dry-hard, hours at 73°F, 3 mils wet film	—	—	—	—	—	8
Set-to-touch time, hours at 73°F, 3 mils wet film	—	—	—	—	—	2
Pot life, hours at 73°F	—	—	—	—	4	—
Consistency, grams	—	—	250	500	150	300

1/ The amount and type of thixotrope shall be selected by the manufacturer to meet all requirements of the general specification and this specification sheet.

2/ GENAMID 2000, Henkel Corporation; UNIREZ 2810, Union Camp; AZAMIDE 600, AZ Products; ANCAMINE 507, Pacific Anchor Chemical Corp.; EPOTUF SF7791, Reichhold Chemical; TriChem 9200, Trimont Chemicals.

3/ VERSAMID 280B75, Henkel Corporation; UNIREZ 2180B75, Union Camp; AZAMIDE 680B75, AZ Products; ANCAMINE 700B75, Pacific Anchor Chemical Corp.; EPOTUF SF7792, Reichhold Chemical; TriChem 9280-B-75, Trimont Chemicals.

4/ 90 percent pure by gas chromatography, density 0.93, refractive index (68°F) = 1.407, IR spectrum characteristic of compound.

5/ 99 percent pure by gas chromatography, density 0.90, refractive index (68°F) = 1.418, IR spectrum characteristic of compound.

6/ Zinc No. 555, American Smelting & Refining; Zinc No. 422, New Jersey Zinc; or Intermediate No. 32 non-gassing, Pacific Smelting Company.

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7/ EPON 828, Shell Chemical Co.; ARALDITE 6010, Ciba-Geigy; DER 331, Dow Chemical Co.; AZEPOXY 128, AZ Products; EPO-TUF 37-140, Reichhold Chemical; EPI-REZ 510, Cleanese; TriChem 728, Trimont Chemicals.

8/ Tests on mixed component shall consist of one volume Component A mixed with four volumes Component B then reduced with 10 percent thinner by volume. Thinner used shall be formula 159 thinner.

Use of alternate ingredients in this formula must have prior approval by the Naval Sea Systems Command. Approval will be based on review of data showing equivalent physical and chemical characteristics to the specified ingredient. It will be necessary to demonstrate that paint made using the ingredient will conform to all requirements of this specification.

Preparing activity:
Navy - SH
(Project 8010-N320)

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